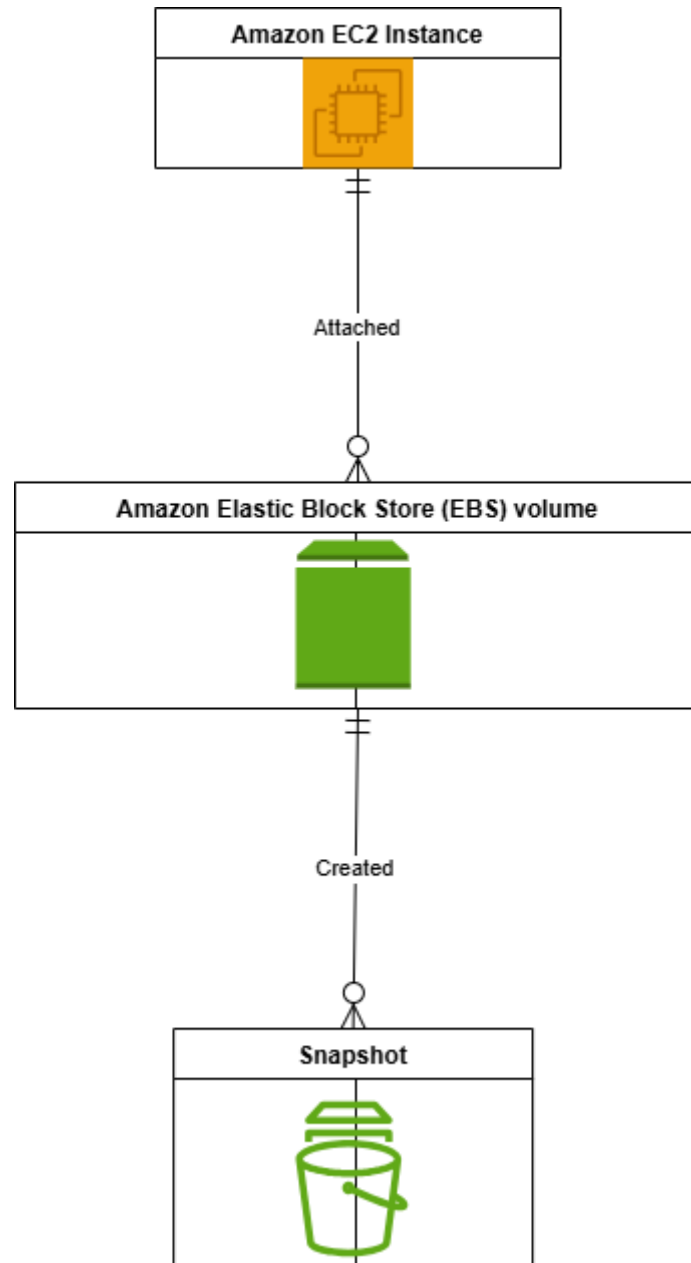
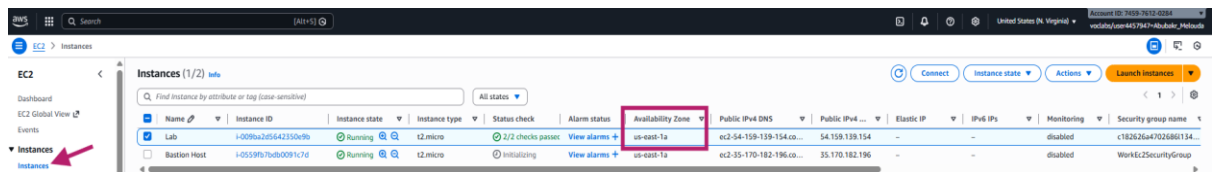
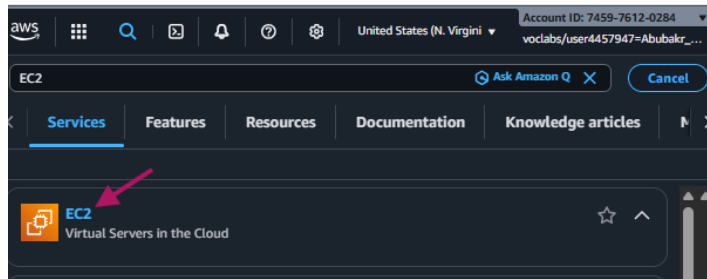


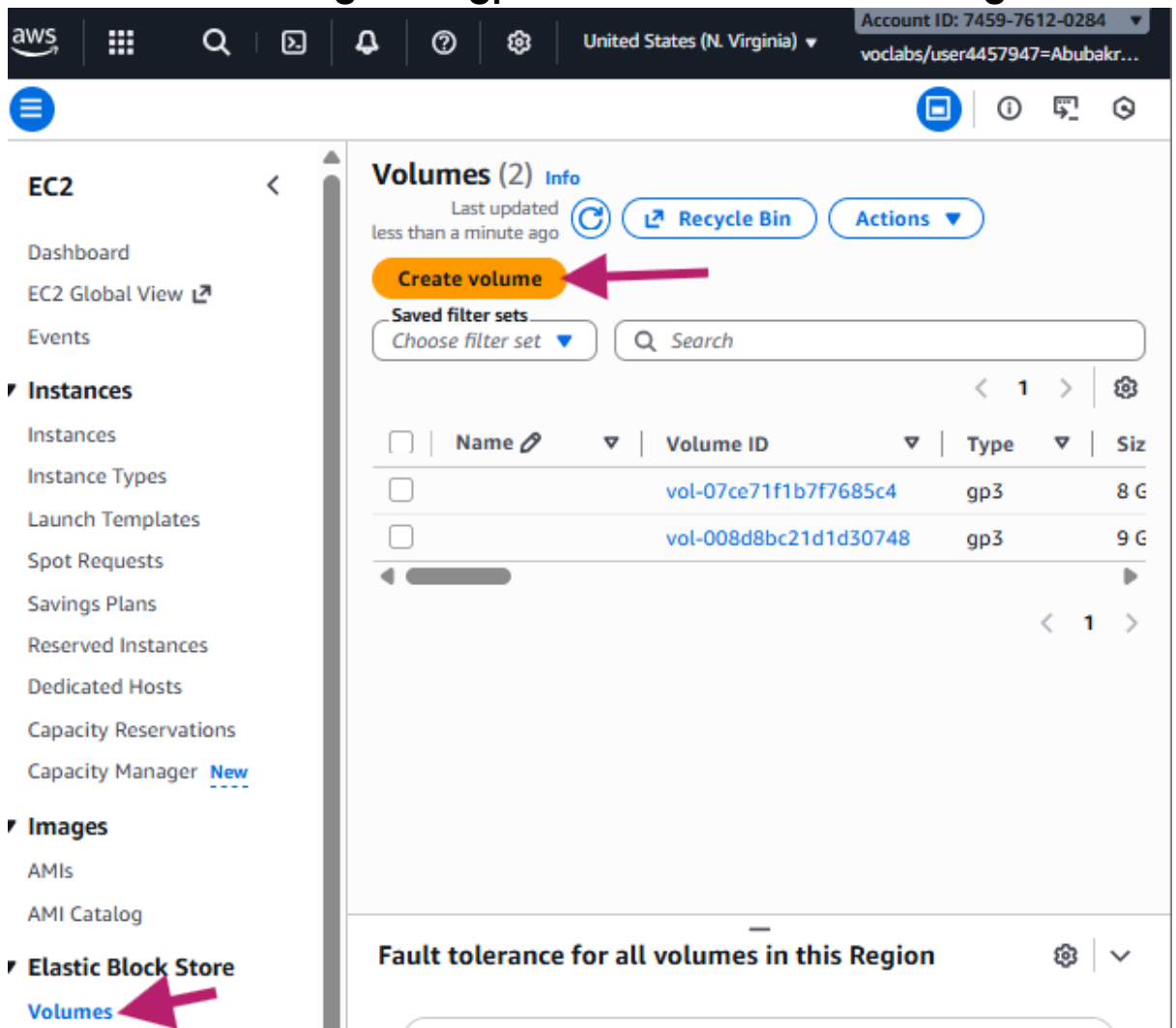
(Module 7 Lab - Working with EBS)



- Screenshot EC2 instance 'Lab' with noted Availability Zone



- Screenshot creating 1 GiB gp2 EBS volume with name tag



aws

United States (N. Virginia) ▼

Account ID: 7459-7612-0284
voclabs/user4457947=Abubakr

EC2 > Volumes > Create volume

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [Info](#)
General Purpose SSD (gp2) ▼

Size (GiB) [Info](#)
1
Min: 1 GiB, Max: 16384 GiB

IOPS [Info](#)
100 / 3000
Baseline of 3 IOPS per GiB with a minimum of 100 IOPS, burstable to 3000 IOPS.

Throughput (MiB/s) [Info](#)
Not applicable

Availability Zone [Info](#)
use1-az2 (us-east-1a) ▼

Snapshot ID - optional [Info](#)
Don't create volume from a snapshot ▼ 

Encryption [Info](#)
Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.
☐ Encrypt this volume

Tags - optional [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
 

Value - optional
 




You can add 49 more tags.

- Screenshot EBS volume successfully created and available

Account ID: 7459-7612-0284
voclabs/user4457947=Abubakr_Melouda

United States (N. Virginia)

EC2

Dashboard
EC2 Global View
Events

Instances

Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager [New](#)

Images

AMIs
AMI Catalog

Elastic Block Store

Volumes
Snapshots
Lifecycle Manager

Network & Security

Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

Load Balancing

Load Balancers
Target Groups
Trust Stores

Auto Scaling

Auto Scaling Groups

Successfully created volume [vol-09c6ffe8996e2525b](#).

Volumes (1/3) Info
Last updated less than a minute ago
Recycle Bin Actions Create volume

Saved filter sets
Choose filter set Search

	Name	Volume ID	Type	Size
☒	My Volume	vol-09c6ffe8996e2525b	gp2	1 GiB
☐		vol-07ce71f1b7f7685c4	gp3	8 GiB
☐		vol-008d8bc21d1d30748	gp3	9 GiB

Volume ID: vol-09c6ffe8996e2525b (My Volume)

Details Status checks Monitoring Tags

Volume ID
vol-09c6ffe8996e2525b (My Volume)

Type
gp2

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

IOPS
100

Fast snapshot restored
No

Created
Wed Jan 07 2026 11:07:21 GMT+0000 (Greenwich Mean Time)

Size
1 GiB

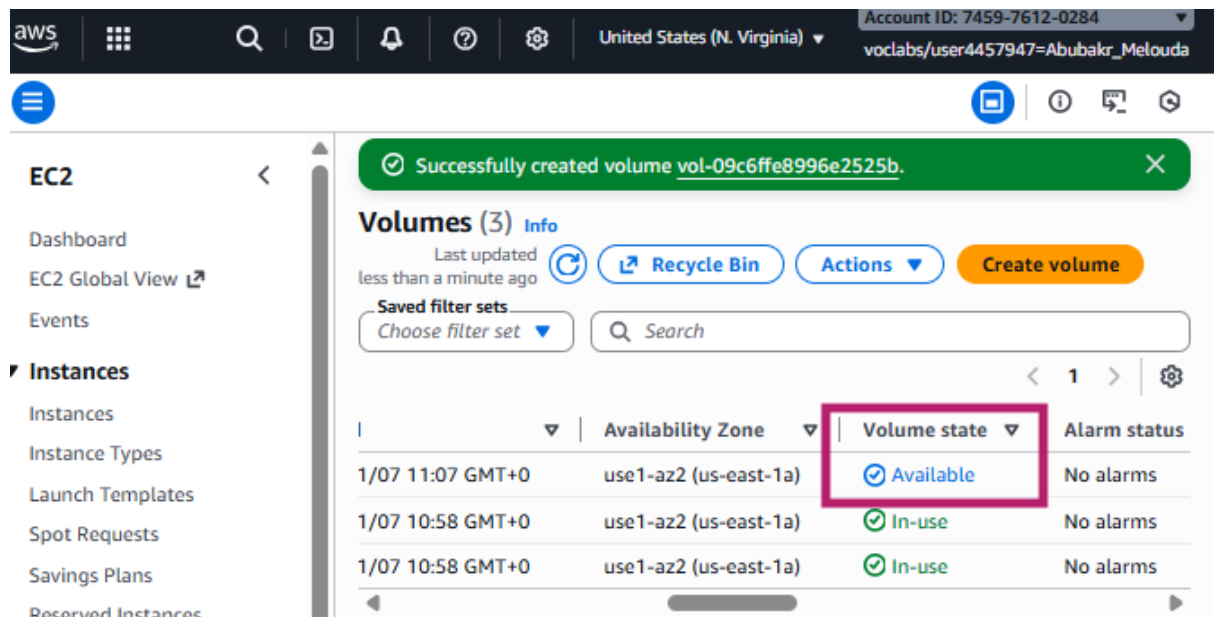
Status check
Okay

Volume state
Creating

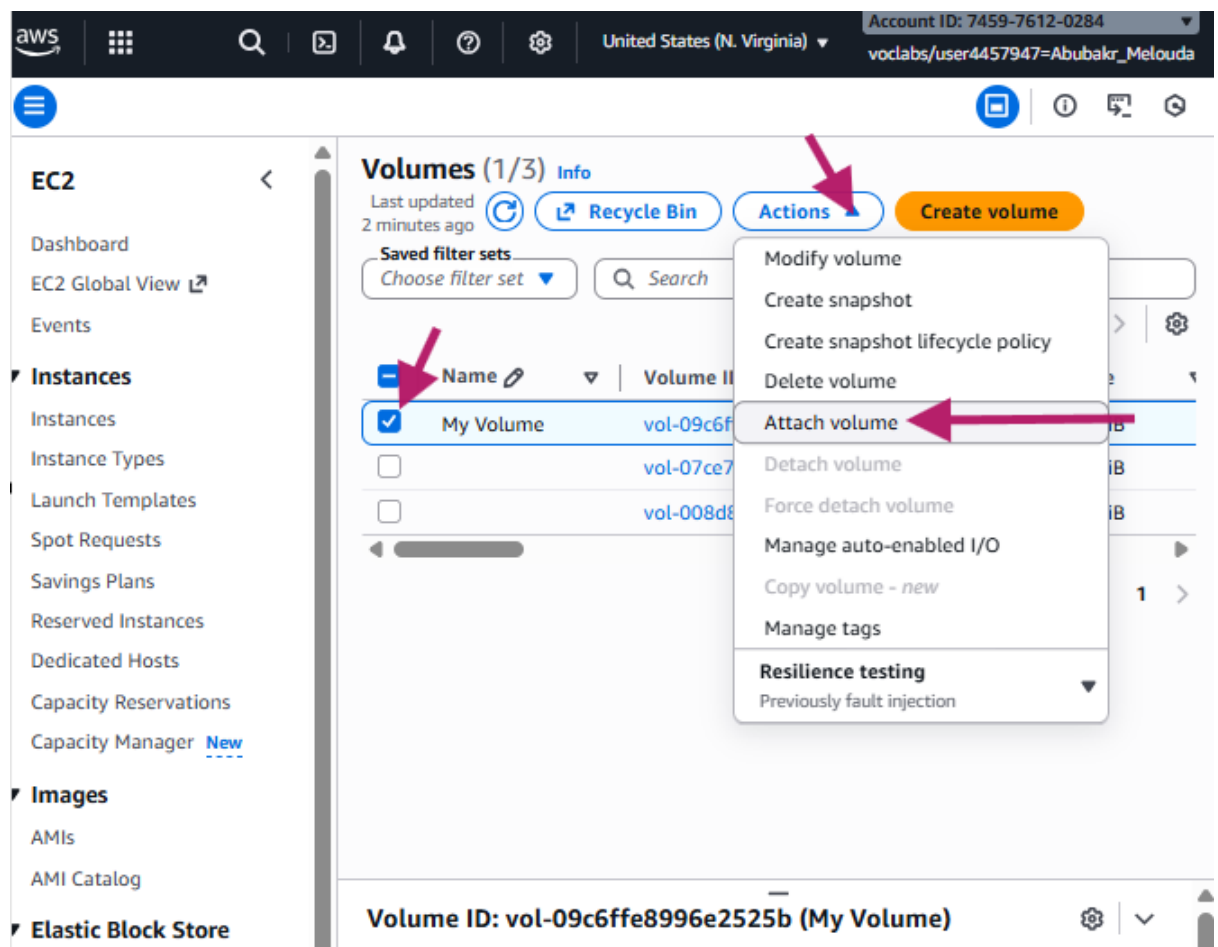
Throughput
-

Availability Zone
use1-az2 (us-east-1a)

Multi-Attach enabled
No



- Screenshot attaching EBS volume to EC2 instance with device /dev/sdf
-



Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID
vol-09c6ffe8996e2525b (My Volume)

Availability Zone
us-east-1a

Instance [Info](#)
i-009ba2d5642350e9b (Lab) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)
/dev/sdf

Recommended device names for Linux: /dev/xvda for root volume. /dev/sd[f-p] for data volumes.

Warning: Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.

[Cancel](#) [Attach volume](#)

- Screenshot EBS volume successfully attached to instance (In-use state)

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created	Availability Zone	Volume state	Alarm status	Attached resources
My Volume	vol-09c6ffe8996e2525b	gp2	1 GiB	100	-	-	-	2026/01/07 11:07 GMT+0	us-east-1a	In-use	No alarms	i-009ba2d5642350e9b (Lab) /dev/sdf (attached)
vol-07c...	vol-07c...	gp3	8 GiB	3000	125	snap-066739f...	-	2026/01/07 10:58 GMT+0	us-east-1a	In-use	No alarms	i-009ba2d5642350e9b (Lab) /dev/xvda (attached)
vol-008...	vol-008...	gp3	9 GiB	3000	125	snap-066739f...	-	2026/01/07 10:58 GMT+0	us-east-1a	In-use	No alarms	i-05589570db0091c7d (Bastion Host) /dev/xvda (attached)

- Screenshot connecting to EC2 instance using EC2 Instance connect

Account ID: 7459-7612-0284
voclabs/user4457947=Abubakr_Melouda

United States (N. Virginia)

EC2 > Instances > i-009ba2d5642350e9b > Connect to instance

Connect

Info

Connect to an instance using the browser-based client.

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

Instance ID

i-009ba2d5642350e9b (Lab)

Connection type

☒ Connect using a Public IP
Connect using a public IPv4 or IPv6 address

☐ Connect using a Private IP
Connect using a private IP address and a VPC endpoint

☒ Public IPv4 address
54.159.139.154

☐ IPv6 address

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

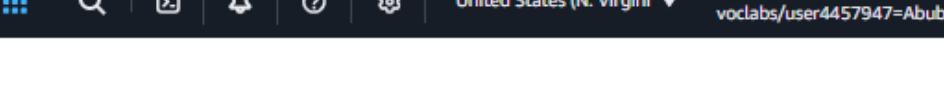
ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

Connect

- **Screenshot EC2 Instance Connect terminal session successfully opened**



Account ID: 7459-7612-0284
voclabs/user4457947=Abubakr_Melouda

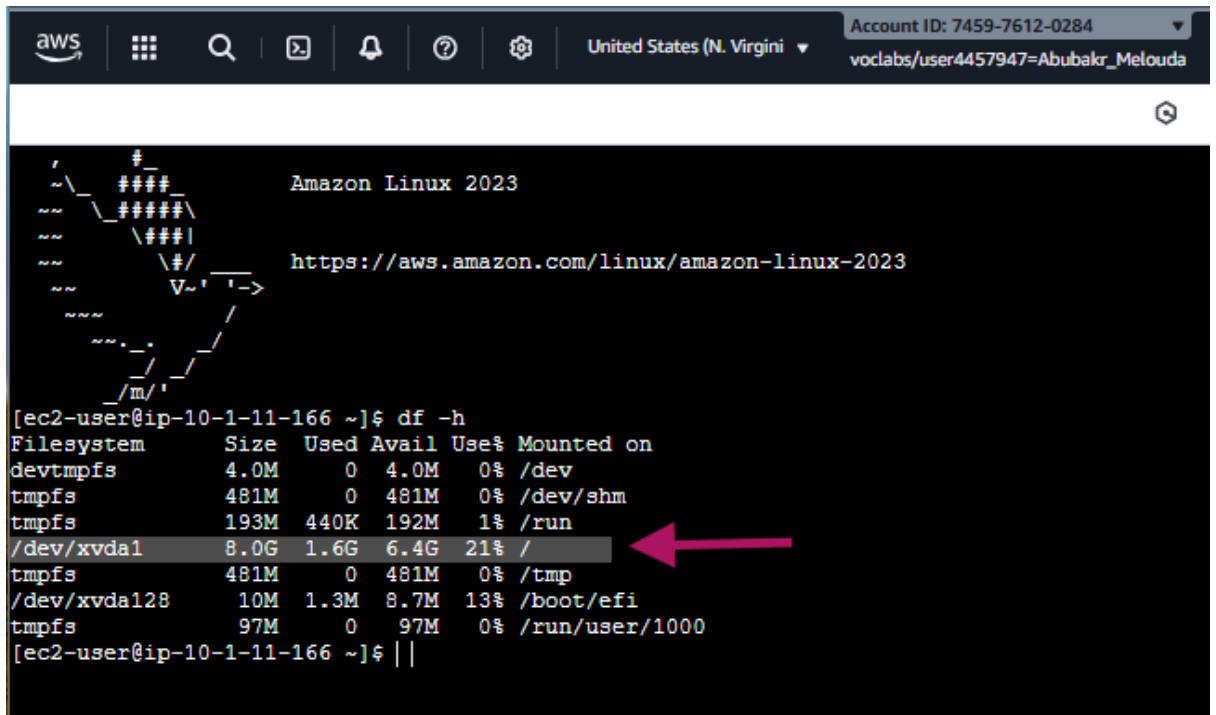
United States (N. Virgini)

Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

[ec2-user@ip-10-1-11-166 ~]\$ df -h

- Screenshot current storage before EBS volume configuration

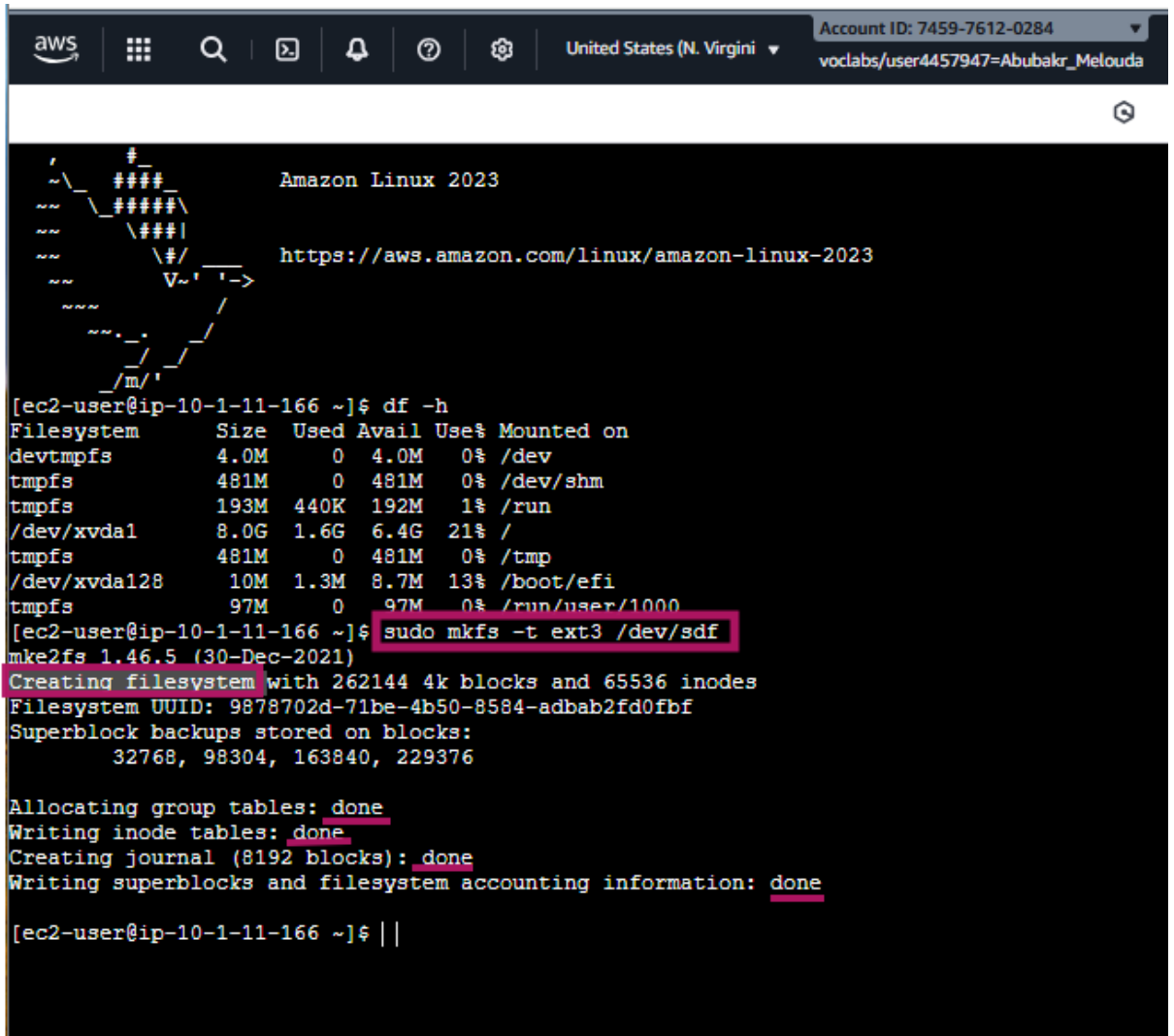


The screenshot shows the AWS Management Console interface. At the top, the account ID is 7459-7612-0284 and the user is voclabs/user4457947=Abubakr_Melouda. The region is United States (N. Virginia). Below the navigation bar, there is a terminal window for an Amazon Linux 2023 instance. The terminal shows the output of the 'df -h' command, which lists the disk space usage for various filesystems. A pink arrow points to the row for '/dev/xvda1', which shows a size of 8.0G, 1.6G used, and 6.4G available.

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-1-11-166 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0  4.0M   0% /dev
tmpfs           481M   0  481M   0% /dev/shm
tmpfs           193M 440K  192M   1% /run
/dev/xvda1       8.0G  1.6G   6.4G  21% /
tmpfs           481M   0  481M   0% /tmp
/dev/xvda128     10M  1.3M   8.7M  13% /boot/efi
tmpfs           97M   0   97M   0% /run/user/1000
[ec2-user@ip-10-1-11-166 ~]$
```

- Screenshot creating ext3 file system on EBS volume



```

aws  [Account ID: 7459-7612-0284] United States (N. Virginia) voclabs/user4457947=Abubakr_Melouda

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-1-11-166 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0  4.0M   0% /dev
tmpfs           481M   0  481M   0% /dev/shm
tmpfs           193M  440K  192M   1% /run
/dev/xvda1       8.0G  1.6G   6.4G  21% /
tmpfs           481M   0  481M   0% /tmp
/dev/xvda128     10M   1.3M   8.7M  13% /boot/efi
tmpfs           97M    0    97M   0% /run/user/1000
[ec2-user@ip-10-1-11-166 ~]$ sudo mkfs -t ext3 /dev/sdf
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: 9878702d-71be-4b50-8584-adbab2fd0fbf
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[ec2-user@ip-10-1-11-166 ~]$ ||

```

- Screenshot fstab configured for automatic volume mounting on boot

```

[ec2-user@ip-10-1-11-166 ~]$ sudo mkdir /mnt/data-store
[ec2-user@ip-10-1-11-166 ~]$ sudo mount /dev/sdf /mnt/data-store
[ec2-user@ip-10-1-11-166 ~]$ echo "/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2"
| sudo tee -a /etc/fstab
/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2
[ec2-user@ip-10-1-11-166 ~]$ cat /etc/fstab
-bash: ecacat: command not found
[ec2-user@ip-10-1-11-166 ~]$ echo "/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2"
| sudo tee -a /etc/fstab
/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2
[ec2-user@ip-10-1-11-166 ~]$ cat /etc/fstab
#
UUID=170fd7a4-f127-46eb-a3be-8b380dc8d94a / xfs defaults,noatime 1 1
UUID=7BBE-6739 /boot/efi vfat defaults,noatime,uid=0,gid=0,umask=0077,sh
ortname=winnt,x-systemd.automount 0 2
/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2
/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2
[ec2-user@ip-10-1-11-166 ~]$ ||

```

- Screenshot EBS volume successfully mounted and visible in storage list

```

[ec2-user@ip-10-1-11-166 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0  4.0M   0% /dev
tmpfs           481M   0  481M   0% /dev/shm
tmpfs           193M  440K  192M   1% /run
/dev/xvda1       8.0G  1.6G   6.4G  21% /
tmpfs           481M   0  481M   0% /tmp
/dev/xvda128     10M   1.3M   8.7M  13% /boot/efi
tmpfs           97M    0   97M   0% /run/user/1000
/dev/xvdf        975M   60K  924M   1% /mnt/data-store
[ec2-user@ip-10-1-11-166 ~]$ ||

```

- Screenshot test file successfully created on EBS volume with content

```

[ec2-user@ip-10-1-11-166 ~]$ sudo sh -c "echo some text has been written > /mnt/data-store/file.txt"
[ec2-user@ip-10-1-11-166 ~]$ cat /mnt/data-store/file.txt
some text has been written
[ec2-user@ip-10-1-11-166 ~]$ ||

```

- Screenshot creating snapshot of EBS volume with name tag

The screenshot shows the AWS Management Console interface for the 'Volumes' section. The top navigation bar includes the AWS logo, a search icon, a notification bell, a help icon, a settings gear, and the region 'United States (N. Virginia)'. The account ID '7459-7612-0284' and the user 'voclabs/user4457947=Abubakr_Melouda' are displayed in the top right corner.

The left sidebar shows the 'EC2' section with a list of navigation links: Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, AMIs, AMI Catalog, Elastic Block Store, and Volumes. The 'Volumes' link is highlighted with a red arrow.

The main content area is titled 'Volumes (1/3)' and includes an 'Info' link. It shows a table of volumes with columns for Name and Volume ID. The first volume, 'My Volume' (vol-09c...), is selected. The 'Actions' dropdown menu is open, showing options like 'Modify volume', 'Create snapshot' (highlighted with a red box), 'Create snapshot lifecycle policy', 'Delete volume', 'Attach volume', 'Detach volume', 'Force detach volume', 'Manage auto-enabled I/O', 'Copy volume - new', 'Manage tags', and 'Resilience testing'. The 'Create volume' button is also highlighted with a red arrow.

Below the table, the 'Volume ID: vol-09c6ffe8996e2525b (My Volume)' is displayed, along with tabs for 'Details', 'Status checks', 'Monitoring', and 'Tags'.

The screenshot shows the AWS Management Console interface for creating an EBS snapshot. The top navigation bar includes the AWS logo, account ID (7459-7612-0284), and region (United States (N. Virginia)). The breadcrumb trail indicates the path: EC2 > Volumes > vol-09c6ffe8996e2525b > Create snapshot.

Create snapshot [Info](#)

Create a point-in-time snapshot to back up the data on an Amazon EBS volume to Amazon S3.

Source volume

Volume ID: [vol-09c6ffe8996e2525b \(My Volume\)](#)

Availability Zone: use1-az2 (us-east-1a)

Snapshot details

Description
Add a description for your snapshot

255 characters maximum.

Encryption [Info](#)
Not encrypted

Tags [Info](#)

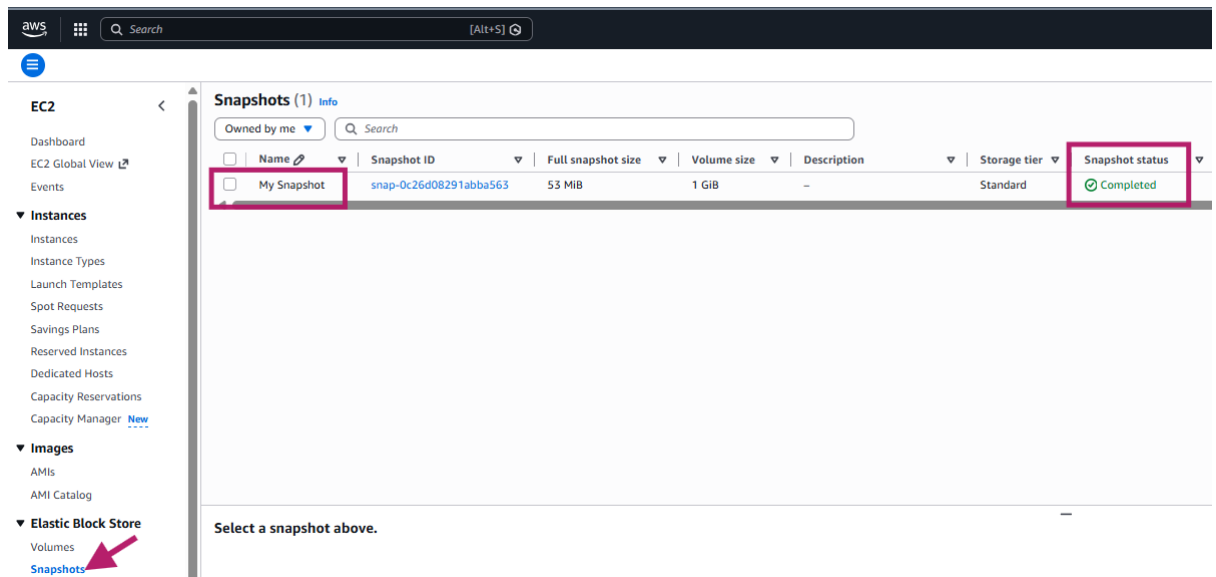
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key:

Value - optional:

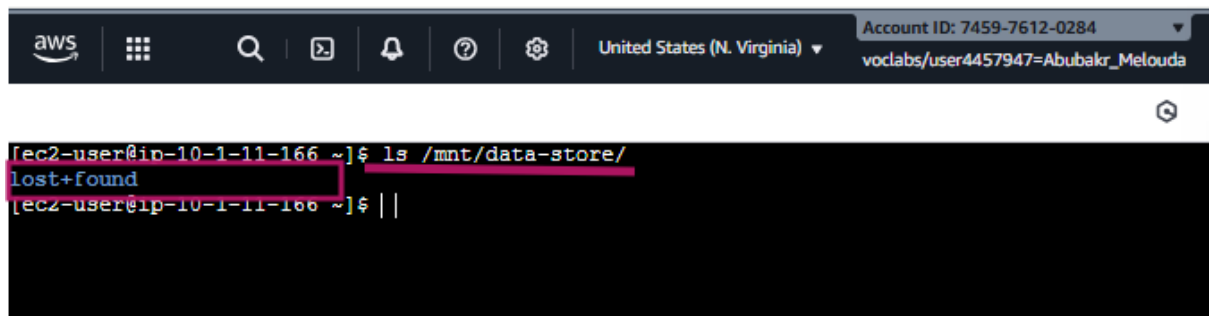
You can add 49 more tags.

- Screenshot EBS snapshot successfully created and completed



- Screenshot test file deleted from volume (simulating data loss)Screenshot

•



- Screenshot creating new volume from snapshot backup
Screenshot

The screenshot displays the AWS Management Console interface. At the top, the header bar includes the AWS logo, navigation icons, the region 'United States (N. Virginia)', and the account ID '7459-7612-0284'. The left-hand navigation pane is expanded to 'Elastic Block Store', where 'Snapshots' is selected. The main content area is titled 'Snapshots (1/1) Info'. It features a table with one entry: 'My Snapshot' with ID 'snap-0c26d08291abba563'. An 'Actions' dropdown menu is open for this snapshot, showing options like 'Create volume from snapshot', 'Create image from snapshot', 'Copy snapshot', 'Launch copy duration calculator', 'Delete snapshot', 'Manage tags', 'Snapshot settings', and 'Archiving'. A red arrow points to the 'Snapshots' link in the sidebar, and another red arrow points to the 'Create volume from snapshot' option in the dropdown menu.

Account ID: 7459-7612-0284
voclabs/user4457947=Abubakr_Melo

United States (N. Virginia)

EC2

Dashboard

EC2 Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager [New](#)

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Snapshots (1/1) Info

Last updated 7 minutes ago

[Recycle Bin](#) [Actions](#) [Create snapshot](#)

Owned by me Search

☒	Name	Snapshot
☒	My Snapshot	snap-0c26d08291abba563

Create volume from snapshot

Create image from snapshot

Copy snapshot

Launch copy duration calculator

Delete snapshot

Manage tags

Snapshot settings

Archiving

Snapshot ID: snap-0c26d08291abba563 (My Snapshot)

[Details](#) [Snapshot settings](#) [Storage tier](#) [Tags](#)

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Snapshot ID
[snap-0c26d08291abba563](#) (My Snapshot)

Volume type [Info](#)
General Purpose SSD (gp3)

Size (GiB) [Info](#)
1
Min: 1 GiB, Max: 65536 GiB.

IOPS [Info](#)
3000
Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) [Info](#)
125
Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

Availability Zone [Info](#)
us-east-1a

Fast snapshot restore [Info](#)
Not enabled for selected snapshot

Volume initialization rate (MiB/s) - new, optional [Info](#)
Specify the rate at which the snapshot blocks are to be downloaded from Amazon S3 to the volume. [Additional costs apply](#)
Enter a value
Min: 100 MiB/s, Max: 300 MiB/s.

Encryption [Info](#)
Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.
☐ Encrypt this volume

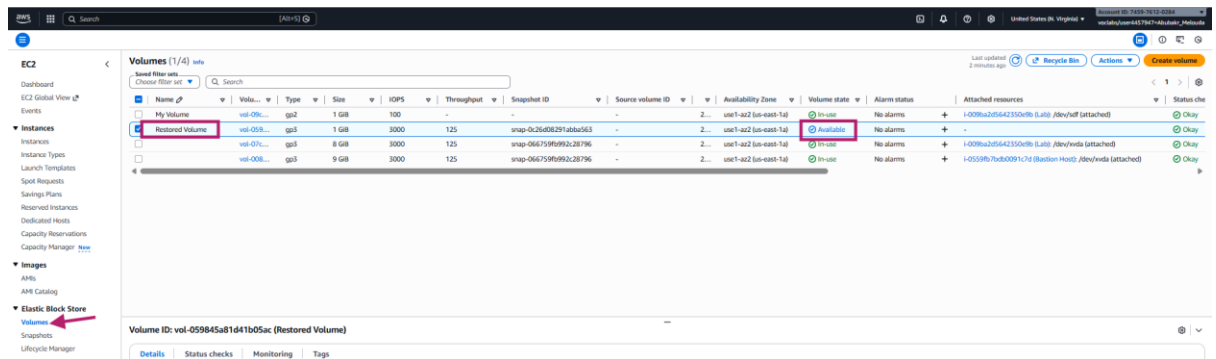
Tags - optional

[Info](#)
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

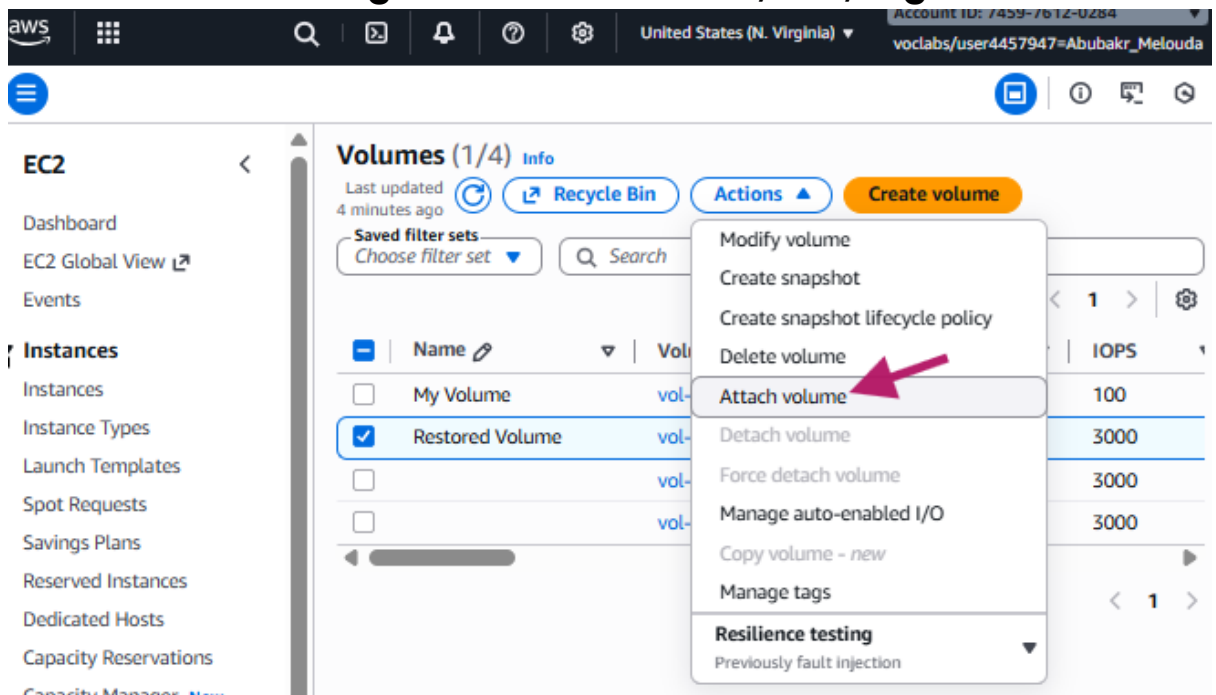
Key	Value - optional	
Name	Restored Volume	Remove

[Add tag](#)

- Screenshot new volume created from snapshot (available state)



- Screenshot attaching restored volume as /dev/sdg device



Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID
vol-059845a81d41b05ac (Restored Volume)

Availability Zone
use1-az2 (us-east-1a)

Instance [Info](#)
i-009ba2d5642350e9b (Lab) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)
/dev/sdg

Recommended device names for Linux: /dev/xvda for root volume. /dev/sd[f-p] for data volumes.

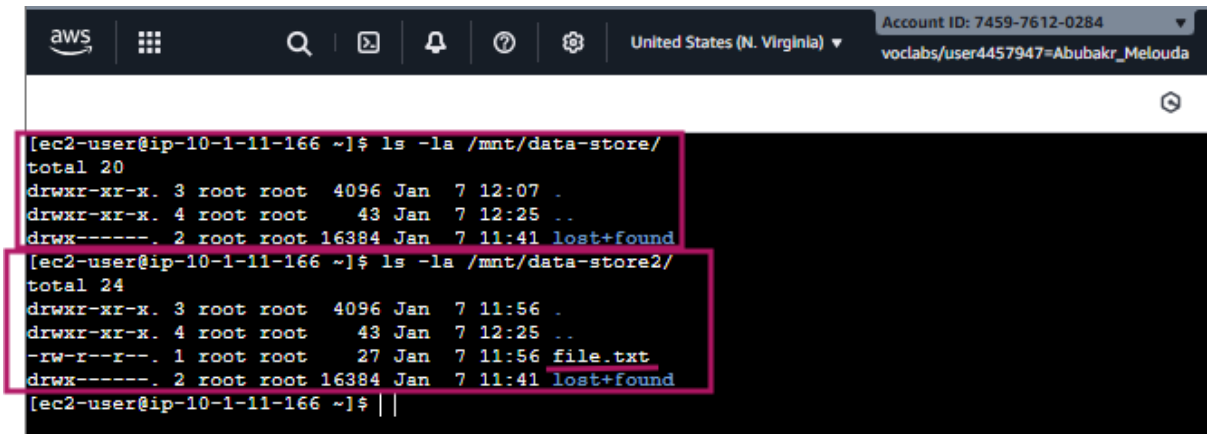
Notification: Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.

[Cancel](#) [Attach volume](#)

- Screenshot data successfully restored from snapshot - file.txt with original content recovered

```
[ec2-user@ip-10-1-11-166 ~]$ sudo mkdir /mnt/data-store2
[ec2-user@ip-10-1-11-166 ~]$ sudo mount /dev/sdg /mnt/data-store2
[ec2-user@ip-10-1-11-166 ~]$ ls /mnt/data-store2/
file.txt  lost+found
[ec2-user@ip-10-1-11-166 ~]$ cat /mnt/data-store2/file.txt
some text has been written
[ec2-user@ip-10-1-11-166 ~]$
```

- Screenshot comparison showing original volume empty, restored volume has data



```
[ec2-user@ip-10-1-11-166 ~]$ ls -la /mnt/data-store/
total 20
drwxr-xr-x. 3 root root 4096 Jan 7 12:07 .
drwxr-xr-x. 4 root root  43 Jan 7 12:25 ..
drwx----- 2 root root 16384 Jan 7 11:41 lost+found

[ec2-user@ip-10-1-11-166 ~]$ ls -la /mnt/data-store2/
total 24
drwxr-xr-x. 3 root root 4096 Jan 7 11:56 .
drwxr-xr-x. 4 root root  43 Jan 7 12:25 ..
-rw-r--r-- 1 root root   27 Jan 7 11:56 file.txt
drwx----- 2 root root 16384 Jan 7 11:41 lost+found
[ec2-user@ip-10-1-11-166 ~]$ |
```

Lab Summary:

Created EBS volume, attached to EC2 instance, created file system, stored data, created snapshot backup, simulated data loss and restored from snapshot.

Key Features:

- EBS volume creation and attachment
- File system creation (ext3)
- Data storage and management
- Snapshot backup creation
- Disaster recovery simulation
- Data restoration from snapshot
- Volume mounting and configuration